

Hejar Shahabi

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 [LinkedIn](#)  [GitHub](#)  [Google Scholar](#)

Professional Summary

Results-driven GIS and Remote Sensing Specialist with over 5 years of experience in spatial analysis, remote sensing, and geospatial data management. Expertise in conducting complex vector and raster-based spatial analyses to support environmental, engineering, planning, and geotechnical projects. Proficient in the full Esri ArcGIS ecosystem (ArcMap, ArcGIS Pro, ArcGIS Online) combined with strong programming skills in Python, ArcPy, SQL, and GEE for automation and workflow optimization. Experienced in processing satellite imagery, airborne and UAV LiDAR, and producing high-quality cartographic products for both desktop and web environments. Strong collaborator with a track record of supporting multidisciplinary teams, maintaining data integrity, and delivering actionable geospatial insights under tight deadlines.

Qualifications

- 5+ years of hands-on experience with ESRI ArcGIS suite (ArcGIS Pro, ArcGIS Desktop 10.x) and QGIS.
- Strong cartographic skills with the ability to produce high-quality 2D maps, figures, charts, and presentation materials from complex data.
- Proficient in creating, converting, and managing GIS data from CAD, tabular, text, and various geospatial formats while maintaining data integrity and QA/QC standards.
- Experience handling field-collected data organizing, validating, analyzing, and mapping under tight deadlines.
- Skilled in geodatabase (PostgreSQL), spatial data integration, publishing, and web-based solutions using ArcGIS Online and ArcGIS Enterprise.
- Strong Python skills for GIS workflow automation, spatial ETL, geoprocessing, and model building.
- Excellent communication and teamwork skills in fast-paced project environments.

Professional Experience

Remote Sensing Analyst - Mitacs Internship

Jun 2024 – Present

Beneva Insurance Company, Québec, Canada

- Developed AI models for high-resolution flood risk mapping using deep learning on geospatial and remote sensing data
- Analyzed aerial imagery to classify building materials (walls, chimneys, garages) for insurance risk modeling
- Developed labeling protocols, annotated datasets, and built supervised and self-supervised AI models for segmentation and object detection

Remote Sensing Analyst

Sep 2022 – Jan 2023

Statistics Canada, Ottawa, Canada

- Implemented deep learning pipelines for solar panel detection in high-res satellite/aerial imagery
- Vectorized outputs in QGIS/Python and prepared data for national renewable energy statistics

Director, Geospatial & Surveying Division

Sep 2017 – Aug 2020

Sotorgsaz Company, Urmia, Iran

- Led GIS/surveying projects, integrated geospatial tech into operations, and ensured data quality/compliance

- Processed spatial data and managed geo-databases for watershed management

Technical Skills

Programming & ML/DL: Python (expert), PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, JavaScript, HTML/CSS

Geospatial & EO: Google Earth Engine (GEE), Geemap, GeoPandas, ArcGIS Pro, ArcMap, QGIS, ENVI, eCognition, ERDAS, SNAP, SARscape

Other: PostgreSQL, Survey 123, AutoCAD/Civil 3D, Microsoft Office

Open-Source Packages (PyPI):

- **GeoPatch** – Patch extraction for RS imagery (>40k downloads) [PyPI](#)
- **PySIC** – ML-based satellite image classification [PyPI](#)
- **TerraTiff** – A lightweight, GDAL-free Python package for reading, writing, and exporting GeoTIFF raster files.

Languages: English (C1 Advanced), Farsi/Azerbaijani/Kurdish (Native), French (Basic), Turkish (Intermediate)

Education

- **PhD in Earth Sciences (Remote Sensing & GeoAI)** 2022–2026
Institut national de la recherche scientifique (INRS), Québec, Canada
- **MSc in Remote Sensing and GIS** 2015–2017
University of Tabriz, Tabriz, Iran
- **BSc in Surveying Engineering** 2010–2013
University College of Omran and Tosseeh, Hamadan, Iran

Awards & Professional Activities

- PBEEE Doctoral Research Scholarship (FRQ) – May 2024–Dec 2025
- INRS PhD Scholarships & Tuition Exemption – 4-year funding (2022-2025)
- 2nd Place, IEEE GRSS 3-Minute Thesis, Canadian Symposium on Remote Sensing (2022)
- Editor's Choice Award, Remote Sensing Journal (2019) – Co-authored

Publications

Top cited papers - over 100 citations:

- **Shahabi, H.**, Jarihani, B., Tavakkoli Piralilou, S., Chittleborough, D., Avand, M., & Ghorbanzadeh, O. (2019).
A semi-automated object-based gully networks detection using different machine learning models: A case study of Bowen catchment, Queensland, Australia.
Sensors, 19(22), 4893.
- **Shahabi, H.**, Rahimzad, M., Tavakkoli Piralilou, S., Ghorbanzadeh, O., Homayouni, S., Blaschke, T., Lim, S., & Ghamisi, P. (2021).
Unsupervised deep learning for landslide detection from multispectral Sentinel-2 imagery.
Remote Sensing, 13(22), 4698.

- **Ghorbanzadeh, O.**, Shahabi, H., Crivellari, A., Homayouni, S., Blaschke, T., & Ghamisi, P. (2022). Landslide detection using deep learning and object-based image analysis. *Landslides*, 19(4), 929–939.
- **Tavakkoli Piralilou, S.**, Shahabi, H., Jarihani, B., Ghorbanzadeh, O., Blaschke, T., Gholamnia, K., Meena, S. R., & Aryal, J. (2019). Landslide detection using multi-scale image segmentation and different machine learning models in the higher Himalayas. *Remote Sensing*, 11(21), 2575.

References

Available upon request.